# One coupling interval, repeated for the whole run

Gas chemistry and aerosol take turns (operator splitting). In each interval $\Delta t_C$ (5 min in Levels 1–2, 15 min in Level 3) the chemistry runs first, its products go to the aerosol model, and the vapors the particles took up or released go back to the gas. PPM marks where particles move between size bins with the piecewise-parabolic method.

### Scenario

- initial gases: limonene, $O_3$, NO, $NO_2$, $SO_2$, CO, $CH_4$ …
- seed aerosol: number, size, sulfate / organic (goes straight to step 3)
- T, RH (constant for the run); site and date (Level 3)
- $\gamma(N_2O_5)$; c* table; number of size bins (40–320)

### 1 Gas-phase chemistry

**Photolysis rates J**
Level 3: TUV-x from latitude, longitude, date and time (at the middle of $\Delta t_C$).
Level 2: constant light (a 30° sun scaled to $J(NO_2) = 5\times10^{-3}\ s^{-1}$, about half).

**Rate constants** for all 2135 reactions, frozen for $\Delta t_C$: MCM expressions, J, the $RO_2$ sum, and $k_{het}(N_2O_5)$ from the particles.

**Stiff ODE solver** JAX
MCM v3.3.1 limonene subset, 725 species (diffrax Kvaerno3, rtol $10^{-3}$), integrated over $\Delta t_C$.

### 2 Hand-off

**Sulfuric acid**
$SO_2 + OH \rightarrow H_2SO_4$, counted over $\Delta t_C$ → a production rate

**Nitrate**
$N_2O_5$ taken up by particles → 2 $NO_3^-$ per $N_2O_5$ → a production rate

**Condensable organics**
~250 products grouped by c* into 15 volatility classes (half decades, $10^{-3}$ to $10^4$ µg $m^{-3}$; c* from Li et al. 2016 or a table) — the VBS gas phase

### 3 Aerosol microphysics

TOMAS sectional model, 60 s micro-steps over $\Delta t_C$ (page 2)

- sources: $H_2SO_4$ vapor; nitrate onto particles PPM
- nucleation ($H_2SO_4$–$H_2O$; off in the presets)
- coagulation
- $H_2SO_4$ condensation PPM
- **SOA partitioning** (vbs-jax): condensation and evaporation per class and size bin PPM
- water uptake (equilibrium)

### 4 Hand-back

- change of each class's vapor → back to the individual species (by gas share when condensing, by particle share when evaporating)
- per-species particle mass → "top contributors"
- Level 3: $NO_2$ reset to 1 ppb (stand-in for emissions)

particle surface area → $k_{het}(N_2O_5)$ and each bin's nitrate share, for the next interval

next interval: gases changed by condensation and evaporation (vapors can react again)

### What the particles carry

- **sulfate**: the seed plus condensed $H_2SO_4$
- **POA**: the seed's non-volatile organic; absorbs SOA
- **SOA**: 15 volatility classes; a per-species tracker shows which products they hold
- **nitrate**: only from $N_2O_5$ uptake (non-volatile)
- **water**: in equilibrium with RH

Not in the model: ammonium nitrate and $NH_3$, $HNO_3$ partitioning, sea salt, dust; dilution, deposition and wall loss.

### The three levels

- **Level 1, Volatility**: no chemistry (steps 1, 2 and 4 are skipped). A surrogate organic with the c* you choose is produced straight into one class; $H_2SO_4$ at a production rate you choose.
- **Level 2, Limonene SOA**: full chemistry, constant light, 6 h.
- **Level 3, Day and night**: TUV-x light over 24 h, $N_2O_5$ uptake ($\gamma$) makes nitrate at night, $NO_2$ reset to 1 ppb every 15 min.

### Clocks and grids

- coupling interval $\Delta t_C$: 300 s (Levels 1–2), 900 s (Level 3)
- aerosol micro-step: 60 s; vbs-jax sub-steps inside it
- size bins: 40 (default), 80, 160 or 320, always 1.7 nm to 17.5 µm at sulfate density (plots: 1.8 nm to 19 µm)
- output: every 10 min (Levels 1–2), 30 min (Level 3)
- all in float64 JAX; one compile per grid and level



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# Inside one aerosol micro-step (60 s)

Step 3 on page 1, in the order the model runs it. Each step changes the size distribution (number and mass of each species in each size bin) before the next one starts.

**1** **Sources.** $H_2SO_4$ made by the chemistry enters the gas at a steady rate. Nitrate from $N_2O_5$ uptake is added to the particles, each size bin getting its share of the uptake (larger surface, larger share); the particles that grew are then moved to the bins that match their new mass PPM.
Then MNFIX MNFIX keeps each bin's mean mass inside the bin. Level 1: the surrogate vapor and $H_2SO_4$ come from your production rates.

**2** **Nucleation.** New 1.7 nm particles at a rate from $H_2SO_4$ and water (Riccobono / Dunne scheme with $NH_3$, ions and organics set to zero, so binary $H_2SO_4$–$H_2O$); as in TOMAS each new particle is 10 % organic by mass. Off in the presets; it needs high $H_2SO_4$ and cold air.
Adaptive sub-steps; MNFIX afterwards MNFIX.

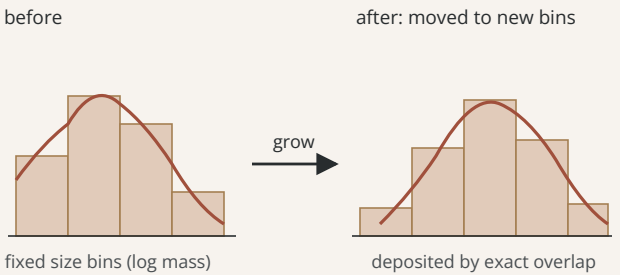
**3** **Coagulation.** Particles collide and stick: number falls, mass stays. 40 bins: the two-moment scheme of Fortran TOMAS (Tzivion, Feingold and Levin 1987); 80–320 bins: the linear discrete method (Simmel et al. 2002).
Ten 6 s forward-Euler sub-steps with the kernel fixed for the 60 s, MNFIX after each MNFIX.

**4** **$H_2SO_4$ condensation.** The vapor condenses onto every bin at the kinetic rate (Fuchs–Sutugin correction in Dahneke's form, accommodation 1); it never evaporates. Particles grow across bin edges PPM.
Flux-form PPM in log-mass with a monotone limiter (Colella and Woodward 1984); MNFIX before and after MNFIX, then particle water is reset.

**5** **SOA partitioning (vbs-jax).** For each volatility class and size bin: condensation where the vapor exceeds the bin's equilibrium vapor, evaporation where it is below. Equilibrium vapor = c* (corrected to the run temperature) × the class's share of the organic mass (Raoult, by mass) × the Kelvin term ($\sigma$ = 0.025 N $m^{-1}$). Seed POA and SOA absorb the vapor; a particle with no organic yet still takes up vapor above its c* (absorbing floor, ASSUMPTIONS AC1). Then each bin's particles move by their own growth or shrinkage PPM.
vbs-jax adaptive sub-steps with a Raoult fixed-point loop; the PPM remap carries each bin's parabolic sub-bin profile and its composition to the bins that match the new masses; organic that would push the top bin off the grid stays in the gas; MNFIX keeps a tiny floor in empty bins MNFIX.

**6** **Water.** Particle water in equilibrium with RH: ammonium-bisulfate water ratio for sulfate and, as a simplification, for organics and nitrate too (organics come out too hygroscopic; ASSUMPTIONS A9).

## What PPM does here




A sectional model keeps particles in fixed size bins. When particles grow, they must be moved to bigger bins without smearing the distribution or losing mass. PPM fits a parabola to the number in each bin (limited, so it never goes negative or overshoots), shifts it by the growth, and deposits it onto the fixed bins. Number and the mass of every species are conserved to round-off.

- **Used for**: $H_2SO_4$ condensation, SOA growth and evaporation, nitrate added by $N_2O_5$ uptake.
- **Before PPM** (tag aaar-stable-2026-10-15): SOA and nitrate were moved by MNFIX, which made the size distribution jagged; with PPM the mean diameter changes by under 0.6 % from 40 to 160 bins.

## MNFIX, still used

TOMAS's bookkeeping fix: if a bin's mean particle mass has left the bin (after sources, nucleation, coagulation, $H_2SO_4$ condensation and the SOA remap), part of the bin moves to the neighbouring bin so number and mass stay consistent. With PPM it only does small corrections; the old SOA and nitrate path (`growth_bin_transfer="mnfix"`) is kept for comparison.

## References for the numerics

Colella, P. and Woodward, P. R.: The Piecewise Parabolic Method (PPM) for gas-dynamical simulations, J. Comput. Phys., 54, 174–201, 1984. · Tzivion, S., Feingold, G. and Levin, Z.: An efficient numerical solution to the stochastic collection equation, J. Atmos. Sci., 44, 3139–3149, 1987. · Simmel, M., Trautmann, T. and Tetzlaff, G.: Numerical solution of the stochastic collection equation – comparison of the Linear Discrete Method with other methods, Atmos. Res., 61, 135–148, 2002. · Zaveri, R. A. et al.: Modeling kinetic partitioning of secondary organic aerosol and size distribution dynamics, Atmos. Chem. Phys., 14, 5153–5181, 2014. PDF · Pankow, J. F.: An absorption model of gas/particle partitioning of organic compounds in the atmosphere, Atmos. Environ., 28, 185–188, 1994. · Li, Y., Pöschl, U. and Shiraiwa, M.: Molecular corridors and parameterizations of volatility, Atmos. Chem. Phys., 16, 3327–3344, 2016. PDF



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